MIL-I-18286A(AS)
14 August 1967
Superseding
MIL-I-18286(Aer)
15 February 1955

MILITARY SPECIFICATION

INDICATOR GROUP, AN/APA-122

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE AND CLASSIFICATION

1.1 Scope - The equipment covered by this specification shall be designed to provide an auxiliary PPI Indicator for use as an additional indicator in those AN/APS-20E Radar installations where there is a requirement for IP-203/APS-20B Indicators in excess of the two normally provided with the AN/APS-20E Radar Set.

1.2 Classification - The equipment covered by this specification shall be of one type and shall consist of the following:

Units	Type Designation	Max. Allow. Size W(in.)	H(in.)	D(in.)	Max. Allow. Weight Lbs.	Applicable Paragraphs
Indicator	IP-203/APS-20B	19-1/2	12-1/4	21-3/16	70	3.3.1
Power Supply	PP- /APA-122	10-1/4	10-1/4	22		3.3.2
Mountings						
Mounting Base	MT-1135/APS-20B	18-3/8	2-3/4	22-3/4	4.5	
Mounting Base	MT- /APA-122				4.0	

1.3 Accessories - The equipment shall operate with the following accessories, however, these accessories are not to be supplied as part of this equipment.

Radar Set	AN/APS-20E
Identification Set	AN/APX-7
Ground Position Indicator	AN/APA-57B
Radar Compensating Assembly	AN/APA-16

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

2.2 Specifications:

Federal Specifications -

QQ-M-151 Metals: General Specification for Inspection of

FSC-5895

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Military Specifications -

JAN-T-152	Treatment, Moisture and Fungus Resistant, of Communications, Electronic and Associated Electrical Equipment, General Process for
JAN-P-106	Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed
JAN-P-658	Packaging and Packing of Electrical Equipment and Spare Parts (Electronic, Electrical and Electromechanical)
MIL-C-172	Cases; Bases, Mounting; and Mounts, Vibration (For use with Electronic Equipment in Aircraft)
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing; Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Limits and Tests; Aircraft Electrical and Electronic Equipment
MIL-P-7788	Plate, Plastic, Cockpit and Interior Controls Lighting
MIL-E-7894	Electric Power, Aircraft Characteristic of
MIL-R-6993(Aer)	Radar Set AN/APS-20E
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-N-18307(Aer)	Nomenclature and Nameplates for Electronic and Associated Equipment
MIL-T-18303(Aer)	Test Procedures

Standards -

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-200	Electron Tubes

Drawings -

Air Force-Navy Aeronautical Drawings

AND 10461	Connections Input Power Airborne Electronic Equipment, 28V DC 115V single phase AC and/or 115/220V three phase AC

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2.3 Availability of Documents -

(1) When requesting specifications, publications and drawings refer to both title and number. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable specifications and publications may be obtained upon application to the Commanding Officer, Naval Air Station, Johnsville, Pennsylvania, Attn: Technical Records Division.

(3) Copies of applicable Bureau of Aeronautics drawings, the U.S. Navy Reliable Tube List may be obtained upon application to the Bureau of Aeronautics, Navy Department, Washington 25, D. C., Attn: Electronics Division.

2.4 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply.

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the Procuring Agency.

(3) Reference Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of Parts and Materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following factors shall govern.

(1) Parts and Materials as approved by Specification MIL-E-5400 shall be given first consideration.

(2) When the contractors can demonstrate that the use of standard AN or JAN parts or materials will not fulfill the design objectives because of size, weight, performance or other reasons, materials and parts shall be used which most nearly meet or exceed the requirements of the respective standard specification.

(3) When a non-standard part is used, the meeting of general material requirements such as non-inflammability, fungus resistance, non-toxicity, etc., and, the meeting of environmental conditions as required by Specification MIL-E-5400 or MIL-T-5422 shall be of prime consideration.

(4) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is to be expended, the assembly shall be constructed so that the unit as a whole will pass material, electrical and environmental tests, without consideration to the quality of the detail parts. Any assembly built to be expended shall be first approved by the Procuring Agency.

3.1.1 Non-Standard Part Approval and Material Approval - Approval for the use of non-standard parts and materials shall be obtained as outlined in Specification MIL-E-5400 except as follows:

(1) Mechanical parts are expected to meet the material requirements of MIL-E-5400, and shall be submitted as non-standard parts only when the material requirements are not met.

(2) Non-standard parts, as defined in Specification MIL-E-5400 of an electrical nature must be approved before preproduction model of the equipment is delivered to the Navy. The request for approval must be forwarded by the contractor before the part is purchased or fabricated. In addition to or in lieu of the NON-Standard Part Data Sheet the contractor is requested to submit itemized lists of the parts, their sources, and the contractor's drawing or purchase specification covering the parts.

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3.1.2 Design Objectives - Minimum size and weight, simplicity of operation and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives in the design of the equipment specified herein. The use of materials, parts and processes other than those required by Specification MIL-E-5400 shall be investigated and when it appears that a substantial reduction in size and weight or improvement in simplicity of design, performance, or reliability can be realized by their use, a request for approval of a non-standard part shall be submitted to the Procuring Agency for consideration, each request should be accompanied by complete supporting information.

3.1.3 Plug-in Assemblies - The use of easily disconnect plug-in assemblies of one or more related circuits shall be investigated with a view towards reduction of size and weight, simplification of operation and maintenance or improvement in performance.

3.1.4 Choice of Tube Types - Electronic tubes used in the equipment shall be chosen in accordance with the requirements of MIL-E-5400, except where there is a counterpart listed in the U. S. Navy Reliable Tube List the reliable version shall be used. The use of any tube not on these acceptable listings must be approved by the Bureau of Aeronautics, however, before consideration for approval, the contractor must submit the following information:

(1) Quantitative data on the electrical advantages and mechanical adequacy of the proposed tube over the most nearly equivalent type on the acceptable listings.

(2) Test data and performance limits of the tube which must be maintained to insure satisfactory and reliable performance in the equipment.

(3) The tube manufacturer shall have received Military or Navy qualification approval for that particular tube.

3.1.5 Workmanship - Workmanship, fabrication and manufacturing processes shall conform to the requirements of MIL-E-5400.

3.2 General Design Requirements - This equipment shall fulfill all applicable design and performance requirements of Specification MIL-E-5400 unless otherwise specified herein.

3.2.1 Standard Conditions - The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests on the equipment:

Temperature	Room ambient 25 C (+ 15°C or -5°C)
Altitude -	Normal ground
Vibration -	None
Humidity -	Room ambient up to 90% relative humidity
Input power	
Voltage -	115 ± 1.0 V AC and/or 27.5 ± 0.5 V DC

3.2.2 Service Conditions - The equipment shall operate satisfactorily under any of the service conditions or reasonable combinations of service conditions as specified in MIL-E-5400 unless modified herein:

3.2.2.1 Temperature and Altitude Combination - The equipment shall not be required to operate at a temperature in excess of +55° C.

3.2.2.2 Vibration - The requirements of paragraph 3.2.20.1.2 of Specification MIL-E-5400 shall not apply as a requirement of this specification.

3.2.3 Total Weight - The total weight of the equipment excluding cables shall not exceed 125 pounds.

3.2.4 Operational Stability - The equipment shall operate with optimum performance for 24 hours, continuously or intermittently, without necessity for readjustment of any controls which are non-accessible during flight.

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3.2.5 Operating Life -

3.2.5.1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 500 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.5.2 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.6 Interconnecting Cabling - This equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded except that a minimum number of the individual wires may be shielded, when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished.

3.2.7 Cables and Connectors - The equipment shall be designed for the use of cables and connectors in accordance with the requirements of Specification MIL-E-5400. However, cables and that portion of connectors which is to be attached to the cables shall not be supplied as part of the equipment.

3.2.8 Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-E-7894 and shall give specified performance from the following power sources with characteristics as defined in MIL-E-7894 having limits as specified herein. The power required shall not exceed the specified amounts.

(1) AC Power Three Phase

(a) 115/200 Volt Variable Frequency

Operating Voltage Limits	108 to 121 Volts
Operation Frequency Limits	380 to 1000 cps
Reduced Performance Voltage Limits	102-108 and 121-124 Volts
Power (3ϕ)	1000 V. A.

(b) 115/200 Volt Constant Frequency

Operating Voltage Limits	108 to 121 Volts
Operating Frequency Limits	380 to 420 cps
Reduced Performance Voltage Limits	102-108 and 121-124 Volts
Power (3ϕ)	50 V. A.

(2) DC Power 28 Volts System

Operating Voltage Limits	24 to 29 Volts
Power	50 Watts

3.2.9 Input Power Connection - Input power connections shall be in accordance with Drawing AND-10461.

3.2.10 Overload Protection - Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses and other protective devices shall be held to a minimum.

3.2.10.1 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage returns within limits.

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3.2.11 Moisture and Fungus Resistant Treatment - Equipments shall be fungus proofed by selection of parts and materials as non-nutrient for fungus or the parts and materials shall be treated prior to their use in the equipment so that overall spraying of the equipment is not necessary. Overall spraying of the equipment in accordance with JAN-T-152 shall be made in the event selection of parts and materials described above is not possible.

3.2.12 Identification and Marking - The equipment shall be identified and marked as follows:

3.2.12.1 Nomenclature and Nameplates - Nomenclature assignment and Nameplate approval for equipment identification shall be in accordance with the requirements of Specification MIL-N-18307(Aer).

3.2.12.2 Special Markings - Any plug-in units and/or other subassemblies that do not carry nameplates shall carry the following special markings as applicable.

(1) The name or acceptable abbreviations of the name of the plug-in or sub-assembly (e.g. IF Strip, Preamplifier, etc.) shall be stamped on the item if space permits.

(2) The letter and number portion of the major component nomenclature, preceded by p/o, shall be stamped on the item (e.g. p/o R-00/APS-00).

(3) The stamping or markings shall meet the requirements of Specification MIL-N-18307(Aer).

3.2.13 Interchangeability - The interchangeability requirements as specified in MIL-E-5400 shall be extended to require interchangeability of units and component parts of this equipment with the latest model of any previously produced equipments. Such interchangeability shall be measured against a model, manufacturing drawings, or other technical information as provided by the Procuring Agency for that purpose. In the event that requirements of MIL-E-5400 conflict with the requirements for interchangeability, the interchangeability requirements shall govern.

3.2.14 Provisions for Maintenance and Field Testing - Provisions for maintenance and field testing shall meet the requirements of Specification MIL-E-5400.

3.2.15 Radio Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.16 Performance - The performance of the AN/APA-122 Indicator Group, when used with the AN/APS-20E Radar Set, shall be equal or superior to performance of the IP-203/APS-20B Range-Azimuth Indicator as specified by Specification MIL-R-6993(Aer).

3.3 Detail Design Requirements -

3.3.1 Indicator, Range-Azimuth, IP-203/APS-20B - Indicator IP-203/APS-20B, provided as a part of Indicator Group AN/APA-122, shall be identical to and completely electrically and mechanically interchangeable with the Indicator IP-203/APS-20B as specified in Specification MIL-R-6993(Aer).

3.3.1.1 Mounting - The Indicator IP-203/APS-20B shall be capable of mounting on Mounting MT-1135/APS-20B. Mounting MT-1135, provided as a part of Indicator Group AN/APA-122, shall be identical to completely interchange with Mounting MT-1135 as specified in Specification MIL-R-6993(Aer).

3.3.2 Power Supply, PP- /APA-122 - The Power Supply, PP- /APA-122, shall be designed to meet the following requirements:

3.3.2.1 Function - The power supply shall be designed to function as a low voltage power supply for one Indicator IP-203/APS-20E. In addition, a servo assembly consisting of a PPI resolver drive assembly and a servo amplifier shall be contained in this unit.

3.3.2.2 Form Factor and Mounting - The maximum size shall not exceed 10-1/4 x 10-1/4 x 22 inches and shall mount on Mounting MT- /APA-122. Mounting MT- /APA-122 shall be so designed to permit easy removal of the power supply unit in aircraft installation and shall meet the requirements of Specification MIL-C-172 for vibration and shock.

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3.3.2.3 Weight with Mounting - The combined weight of PP- /APA-122 and MT- /APA-122 shall not exceed 50 pounds.

3.3.2.4 Power Outputs - The Power Supply shall furnish all power required by the IP-203/APS-20B indicator except that the high voltage for the indicator CRT shall be provided by the indicator. D.C. voltages to the AN/APA-57 N-S, E-W output potentiometers shall be available at an output receptacle.

3.3.2.5 Cooling - The equipment shall be cooled by means of an AC blower.

3.3.2.6 Output Voltages - In addition to the distribution AC power and +28 volts d.c., this unit shall provide the following output voltages:

	Output Voltage (volts)	Voltage Tolerance (volts)	Maximum Ripple (volts, rms.)	Maximum Continuous Current (ma.)
Regulated	+300	± 2	0.050	100
Regulated	+152	± 2	0.050	250
Regulated	-152	± 2	0.050	250
Unregulated	+270	± 15	1.000	250

3.3.2.7 Servo Assembly - The servo assembly shall consist of a PPI resolver drive assembly and a servo amplifier. The servo assembly shall have the following characteristics:

(1) Operation at antenna speeds up to 30 rpm.

(2) Maximum dynamic error of 20 minutes at antenna speed up to 30 rpm.

(3) Maximum static error 15 minutes.

(4) Resolution accuracy 10 minutes.

3.3.2.7.1 Servo Amplifier - For maximum reliability the servo amplifier shall be of the magnetic type.

3.3.2.7.2 Synchro Input - The equipment shall accept antenna synchro signals from the AS-407/APS-20B or the OA-492A/APS-20B antennas.

3.3.2.7.3 PPI Sweep Output - The equipment shall provide north stabilized PPI sweep outputs to the IP-203/APS-20B indicator.

3.3.2.8 Power Switching - All input power to the equipment shall be controlled by switches on the equipment front panel.

3.3.2.9 Voltage Test Points - Test points shall be provided on the equipment front panel to permit ready measurement of the regulated output voltages.

3.3.2.10 Maintenance Adjustments - Maintenance adjustments shall be readily accessible and shall perform the following functions:

(1) Voltage adjustments (regulated output voltages)

(2) Servo amplifier gain control

(3) Servo amplifier balance control

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4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests - The testing and inspection of aircraft electronic equipment shall be classified as follows:

(1) Preproduction Tests - Preproduction tests are those tests conducted on samples which are representative of the production equipment to be supplied under the contract, for the purpose of determining that the equipment meets all the requirements of the applicable specifications. Preproduction Tests shall consist of the following two tests.

(a) Contractor Demonstration Tests - Contractor Demonstration Tests are accomplished under the responsibility of the contractor on a sample, representative of the production equipment, to demonstrate that the equipment complies with all the requirements of the applicable specifications.

(b) Service Approval Tests - Service Approval Tests are accomplished under the responsibility of the procuring service on a sample representative of the production equipment, to determine that the equipment meets all the requirements of the specification. These tests may be accomplished by having the procuring service supervise the Contractor Demonstration Tests.

(2) Inspection Tests - Inspection Tests are those tests accomplished on equipment submitted for acceptance under the contract. Inspection Tests shall consist of the following.

(a) Individual Tests - Individual Tests are those tests conducted on each equipment submitted for acceptance under the contract.

(b) Sampling Tests - Sampling Tests are those tests accomplished on a quantity of equipment as approved by the Bureau of Aeronautics and which have previously passed the Individual Tests.

4.2 Preproduction Tests -

4.2.1 Contractor Demonstration Tests - The contractor shall conduct tests on one or more sample equipments as necessary to determine that the design of the equipment as proposed by the contractor will meet the requirements of all applicable specifications. Contractor Demonstration Tests shall be conducted in accordance with the approved preproduction test procedure. The data obtained by the contractor in conducting these tests shall be included with the design data submitted with the Service Approval Test sample. The Inspector of Naval Material and the Bureau of Aeronautics shall be advised when tests are to be conducted so that the Bureau of Aeronautics may designate a representative to witness or supervise the test when so desired. When tests are supervised the Bureau of Aeronautics may decide to omit certain portions of the Service Approval Tests which will duplicate the Contractor Demonstration Tests; however, in conducting Service Approval Tests the Bureau of Aeronautics may elect to repeat any test previously conducted by the contractor, when deemed necessary. Contractors not having laboratory facilities to satisfactorily conduct all tests shall obtain either the services of a commercial testing laboratory or written approval from the Bureau of Aeronautics to omit that portion of the tests which cannot be conducted.

4.2.2 Service Approval Tests - The Service Approval sample shall consist of one equipment representative of the production equipment to be supplied under the contract. This equipment shall be delivered by the contractor to the destination specified by the Bureau of Aeronautics. The sampling procedure outlined in Specification MIL-P-7788 shall apply for plastic lighting plates.

4.2.2.1 Accessory Materials required - In addition to the complete equipment submitted, the contractor shall submit the accessory material and design data as specified in Specification MIL-E-5400.

4.2.3 Scope of Tests - Preproduction Tests shall include all tests deemed necessary to determine that the model meets all the requirements of this specification and the contract. Preproduction Tests shall include testing in accordance with Specification MIL-T-5422 and Specification MIL-I-6181. The testing of plastic lighting plates shall be in accordance with Specification MIL-P-7788. The design and test data supplied by the contractor shall indicate the equipment's physical and electrical characteristics and establish that the equipment meets all the requirements of this specification. The test data shall include data obtained by the contractor in testing to MIL-T-5422.

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4.2.4 Preproduction Sample Approval - Approval of the preproduction sample shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No additional equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of any equipments prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample will be returned to the contractor for his use in the fabrication and testing of the equipment to be submitted for acceptance under the contract. The preproduction sample shall not be considered as one of the equipments under the contract, however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Inspection Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having laboratory testing facilities satisfactory to the Government Inspector shall engage the services of a commercial testing laboratory acceptable to the Bureau of Aeronautics. The contractor shall furnish test reports showing quantitative results for all inspection tests required by this specification, and signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. Inspection Tests shall consist of the following:

(1) Individual Tests - Individual tests shall be conducted on each equipment submitted for acceptance under the contract.

(2) Sampling Tests - Sampling Tests shall be conducted on a quantity of equipment as specified in paragraph 4.3.2.1. Equipments for sampling tests shall be selected by the Government Inspector after the equipment has passed the Individual Tests.

4.3.1 Individual Tests - Each equipment shall be examined carefully to determine compliance with the requirements of this specification with respect to workmanship. In addition, the equipment shall be given an operational check, with sufficient data recorded to assure satisfactory operation.

4.3.1.1 Scope of Test - As a minimum each equipment shall be tested to determine compliance with the following:

(1) Alignment of Azimuth System for proper performance when utilized with an AN/APS-20E Radar Set.

(2) Azimuth Accuracy

(3) Accuracy of Range Marks Spacing

(4) Accuracy of Range Strobe

(5) Accuracy of Bearing Cursor

(6) Sweep speeds, lengths and linearity

(7) Stability of Regulated Output Voltages of Power Supply

(8) Ripple Magnitude on Regulated Output Voltages

(9) Overall Accuracy of the PPI Servo System

4.3.2 Sampling Tests - Sampling Tests shall be conducted under the close supervision of the Government Inspector and shall include a performance check at ambient laboratory conditions. This test may also include any of the other preproduction tests the Inspector deems necessary to determine conformance of the equipment with the requirements of this specification. These Tests shall include also a careful examination of the way the equipment is packaged and packed to determine conformance with the packaging and packing requirements of this specification or the contract. Sampling Tests shall be conducted on complete equipments. If individual units of the equipments are individually tested and shipped then the samples selected for Sampling Tests shall consist of all the individual units necessary to compose a complete operating equipment.

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4.3.2.1 Sampling Test Schedule - Sampling Tests shall be performed on the following quantity of equipment unless otherwise stated in the contract or approved in writing by the procuring service.

Quantity of Equipment	Quantity of Equipment to be Tested
First 10 equipments	1
Next 10 to 100	2
Next 100 to 200	1
Over 200	One for each 200

4.3.2.2 Scope of Test - As a minimum, each equipment selected for sample testing shall be tested to determine compliance with the items listed under paragraph 4.3.1.1 under the environmental conditions specified in Specification MIL-T-5422 as modified herein.

4.3.2.3 Equipment Failure - Should an equipment fail to pass any part of the Sampling Test, the following corrective action shall be taken by the contractor.

(1) Determine the cause of the failure.

(2) Determine if the failure is an isolated case or a design defect.

(3) If a design defect, suggest corrective action for Bureau of Aeronautics' approval.

(4) Include a test in the Individual Test to check all equipments for these requirements until reasonable assurance can be obtained that the defect has been satisfactorily corrected.

4.4 Test Procedures - The procedures and methods for conducting Pre-production Tests and Inspection Tests shall be prepared by the contractor and sent to the Bureau of Aeronautics for approval. The right is reserved by the Bureau of Aeronautics or the Naval Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Use MIL-T-18303 as a guide for preparation of Test procedures.

4.5 Pre-Submission Testing - Prior to the submission of any model for Preproduction Testing, the contractor shall furnish facilities for and shall conduct tests as required by paragraph 4.2.1. In addition, the proposed test procedure shall be submitted in sufficient time to permit its review by the Bureau of Aeronautics and all necessary revisions by the contractor prior to the start of any tests.

4.6 Pre-production Sample Approval - Approval of the preproduction sample shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No additional equipments shall be delivered prior to the approval of the pre-production sample. Prefabrication of any equipments prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample will be returned to the contractor for his use in the fabrication and testing of the equipment to be submitted for acceptance under the contract. The preproduction sample shall not be considered as one of the equipments under the contract, however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be packaged, packed and marked for shipment in accordance with the requirements of Specification JAN-P-658 and MIL-STD-129.

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6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 volts dc and 115 ± 1 volts ac. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

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